



1. Raspberry Pi OS
2. Python 3
3. Required Python libraries:
 - Adafruit_DHT
 - Adafruit_BMP280
 - py-sds011
 - RPi.GPIO
 - numpy
 - pandas
 - matplotlib

Section 3: Assembly Instructions

To assemble your weather station, follow these steps:

1. Mount the Raspberry Pi 4 Model B board in the weatherproof enclosure.
2. Connect the DHT22, BMP280, SDS011, anemometer, and rain gauge sensors to the breadboard.
3. Use jumper wires to connect the sensors to the Raspberry Pi GPIO pins as follows:

DHT22:

Connect the DHT22 data pin to GPIO4 (pin 7) on the Raspberry Pi. Connect the DHT22 VCC (3.3V) and GND pins to the Raspberry Pi's respective 3.3V and GND pins.

BMP280:

Connect the BMP280 SDA pin to GPIO2 (pin 3) on the Raspberry Pi. Connect the BMP280 SCL pin to GPIO3 (pin 5) on the Raspberry Pi. Connect the BMP280 VCC (3.3V) and GND pins to the Raspberry Pi's respective 3.3V and GND pins.

SDS011:

Connect the SDS011 TX pin to GPIO14 (pin 8) on the Raspberry Pi. Connect the SDS011 RX pin to GPIO15 (pin 10) on the Raspberry Pi. Connect the SDS011 5V and GND pins to the Raspberry Pi's respective 5V and GND pins.

Anemometer:

Connect the anemometer data pin to GPIO17 (pin 11) on the Raspberry Pi. Connect the anemometer VCC (3.3V)

and GND pins to the Raspberry Pi's respective 3.3V and GND pins.

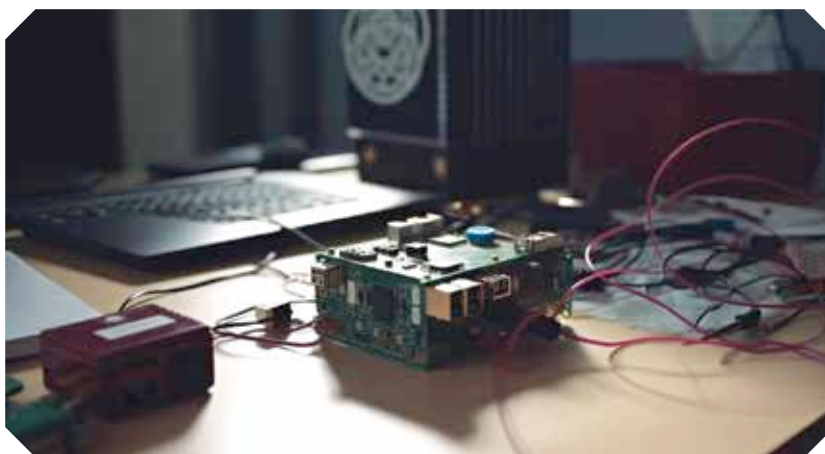
Rain gauge:

Connect the rain gauge data pin to GPIO18 (pin 12) on the Raspberry Pi. Connect the rain gauge VCC (3.3V) and GND pins to the Raspberry Pi's respective 3.3V and GND pins.

4. Connect the required resistors (refer to the circuit diagram in Section 4).
5. Connect the power supply to the Raspberry Pi.

Section 4: Circuit Diagram

Please refer to this circuit diagram for wiring connections:



Section 5: Programming Code

You can use Python 3 to program your weather station. First, install the necessary libraries by running the following commands:

```
pip3 install Adafruit_DHT
Adafruit_BMP280 py-sds011
RPi.GPIO numpy pandas matplotlib
```

Next, create a Python script (e.g., weather_station.py) with the following code snippet:

```
import time
import Adafruit_DHT
import Adafruit_BMP280
```

```
import py_sds011
import RPi.GPIO as GPIO
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
# Initialize DHT22
DHT_SENSOR = Adafruit_DHT.
DHT22
DHT_PIN = 4
```

```
# Initialize BMP280
bmp280 = Adafruit_BMP280.
Adafruit_BMP280(busnum=1)
```

```
# Initialize SDS011
sds011 = py_sds011.
```

```
SDS011("/dev/ttyS0", use_
query_mode=True)
sds011.set_working_
period(0) # Continuous
mode
```

```
# Initialize anemometer and
rain gauge
ANEMOMETER_PIN = 17
RAIN_GAUGE_PIN = 18
GPIO.setmode(GPIO.BCM)
GPIO.setup(ANEMOMETER_PIN,
GPIO.IN, pull_up_down=GPIO.
PUD_UP)
GPIO.setup(RAIN_GAUGE_PIN,
GPIO.IN, pull_up_down=GPIO.
PUD_UP)
```

```
wind_count = 0
```